Predicting the Results of Professional Tennis Matches with Machine Learning

Sergio de Carvalho

Introduction

Predicting the outcome of professional sporting events is a long-standing challenge with broad applications from athletic training to broadcast media engagement. Tennis offers a unique setting for machine learning because results are determined from one-to-one matches, with fewer confounding variables than team sports. Traditional match prediction relies heavily on official rankings that prioritize cumulative participation and tournament prestige over absolute strength of opposition and recent form.

This project establishes a machine learning framework to predict professional tennis match outcomes strictly using pre-match data. Using an extensive tabular dataset spanning over five decades of matches from the Association of Tennis Professionals (ATP), the project focuses on domain-driven feature engineering. Central to the methodology is the integration of dynamic "enhanced covariates"—such as Elo and Glicko **[1]** rating systems—computed strictly in chronological order to eliminate data leakage.

Methodology

- Aggregated historical records from 189,373 matches across tournaments from 1967 to 2024, covering 7,477 players
- Enriched match records with player information: age, height, dominant playing hand, current ATP rank and ranking points
- Calculated historical metrics for each player in strict chronological order: highest rank, total matches played and win rates (career and last 12 months), head-to-head wins
- Maintained alternative rankings based on dynamic paired comparison models (Elo and Glicko) that estimate the abilities of players considering the strength of opponents, including *weighted* Elo and Glicko variants that account for the margin of victory (number of games won in a match) **[2]**
- Considered results on specific court surfaces separately (grass, clay, hard courts) to compute additional *per-surface* metrics such as win rates, head-to-head counts, Elo and Glicko rankings **[3]**
- Trained and tuned six supervised classifiers: Random Forest, XGBoost, CatBoost, Calibrated Linear C-Support Vector Classifier (C-SVC), Nyström kernel approximation with SGD (RBF mapping), and a Multi-Layer Perceptron

Evaluation and Results

- Prediction accuracy fluctuates noticeably by era (Figure 1), tournament level (Figure 2), and match format (matches in best-of-five sets format are easier to predict than best-of-three)
- All models converge near a 70% accuracy ceiling in line with results in the literature **[4]**, outperforming the standalone ATP ranking baseline (66.24%) and overall Elo (68.31%)

Model	Accuracy	Precision	Recall
CatBoost	69.97 %	69.98 %	69.97 %
XGBoost	69.85 %	69.85 %	69.85 %
Multi-Layer Perceptron	69.53 %	69.54 %	69.53 %
Random Forest	69.51 %	69.51 %	69.51 %
Nyström + SGD	69.23 %	69.27 %	69.23 %
Linear C-SVC	68.98 %	68.98 %	68.98 %

Table 1: Overall accuracy, precision and recall of all models.

Analysis

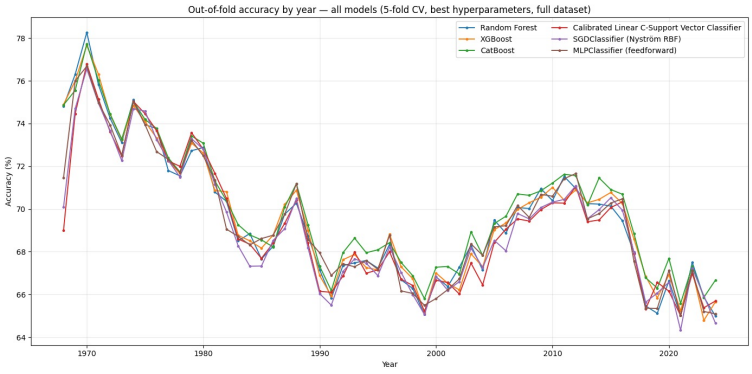


Figure 1: Overall accuracy of predictions from all models, by year.

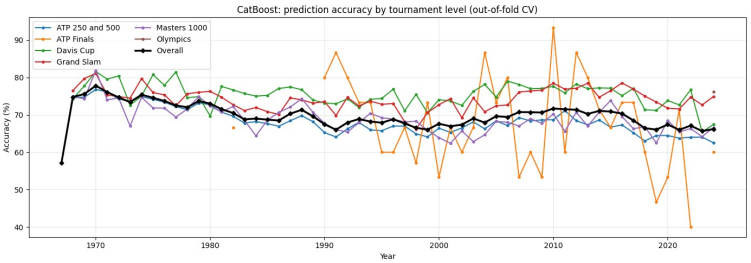


Figure 2: Accuracy of CatBoost model by year and tournament level.

References

[1] Glickman, M.E. (1999). "Parameter estimation in large dynamic paired comparison experiments". In: *Applied Statistics* 48.3, pp. 377-394.

[2] Angelini, G., V. Candila, and L. De Angelis (2022). "Weighted Elo rating for tennis match predictions". In: *European Journal of Operational Research* 297.1, pp. 120-132.

[3] Williams, L.V. et al. (2021). "How well do Elo-based ratings predict professional tennis matches?" In: *Journal of Quantitative Analysis in Sports* 17.2, pp. 91-105.

[4] Wilkens, S. (2021). "Sports prediction and betting models in the machine learning age: The case of tennis". In: *Journal of Sports Analytics* 7.2, pp. 99-117.

Discussion and Future Work

- Predicting professional tennis matches remains challenging, in part due to factors that are not fully captured in historical records: weather conditions, variations in court pace, the mental and physical state of players, tournament schedules, and situational pressure
- Minimal performance differences among algorithms confirm that feature engineering quality—rather than model architectural choice—drives predictive power
- Future work should expand the scope to include the Women's Tennis Association (WTA) circuits as well as lower-tier events (Challengers, Futures, and qualifying rounds)
- Further gains can be pursued by incorporating court speed indexes, indoor/outdoor court status, weather conditions, home advantage, fatigue factors (e.g. total time on court, rest days), as well as match-level metrics (first-serve percentages, aces, double faults, break points saved/won)